Looking at the plot of loss against epochs we are confident that we are not overfitting as both the training and validation losses are trending down closely similar to the accuracy plot above. It also shows that the loss will likely go lower if we increase the epoch size.

Prediction

Now the time has come to reach the final and most interesting stage of predicting values. First, we need to perform pre-processing on the test data. It is similar to that as we did in the training and validation set. Perform the following steps:

1. We first read and convert the top 10 images in our test set to a list of array. It is important to note that the y_test will be empty since the test set has no label.
2. Next, we convert the list of the array to a single numpy array.
3. Finally, we create a test ImageDataGenerator and perform normalisation only.
   Do not perform data augmentation since it is for testing purposes only.

Now we will create a simple iteration loop (for loop), that iterates over the images from the generator to make predictions. Finally, we will plot the results.

1. Create an empty list to hold the labels that are going to be predicted.
2. Set the figure size of the images that we are going to plot.
3. We predict the particular image provided by the ImageDataGenerator from the test dataset by calling the .predict ( ) method on our trained model.
4. The pred variable is a probability gives the probability of the testing image to be a dog. Since we gave dogs a label of one, a high probability or at least greater than average 0.5 means that our model is very confident that the tested image is of a Dog. Else, it is a cat. Hence, we can create an if -else statement that adds the string 'Dog' if the probability is greater than 0.5 else it adds 'cat' to the text_label.
5. Now we add a subplot so we can plot multiple images.
6. Finally, we add the predicted class as a title to the image plot and plot the image.

We are done with this model. Next, we introduce the concept of transfer learning to increase the accuracy of the same model to about 95%.